

Part 1a: Navigating public databases and resources

The purpose of Part 1 is to learn how to navigate and use public databases and resources to understand the inheritance pattern, genomic architecture, and pathogenesis pathway of genetic diseases.

IMPORTANT:

- Your group is assigned to a gene/variant of interest (**G/VOI**) from a genetic disease (that you have selected as a group).
- Complete and submit Lab 1 individually using this gene/variant of interest and disease.
- You may work together as a group to verify the information you've collected across the databases.

Here is a table of the resources we will be using in Lab 1. This is a non-comprehensive table of publicly available resources, but we will use the following to get started.

Database	Purpose
UCSC Genome Browser	Provides genomic and epigenomic data, including visualization tools for genome annotations.
OMIM	A database of human genes and genetic disorders. Use this to investigate gene-disease relationships and find information on genetic disorders.
GeneCards	Integrates information about genes, their functions, expression, and and relevance to diseases.
GeneReviews	Expert-authored summaries on genetic conditions, including diagnostic and treatment guidelines.

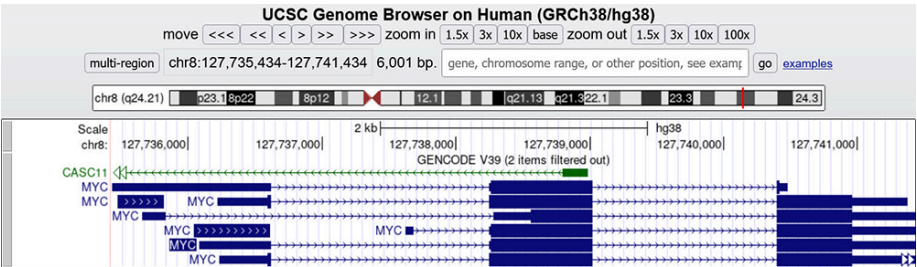
Exercise 1: UCSC Genome Browser

Genome browsers *integrate genomic sequence and annotation data* from different sources and provide a platform to search, browse, retrieve, and analyze genomic data. They differ from ordinary biological databases in that they display data in a graphical format, with genome coordinates on one axis with annotations or space-filling graphics to show analyses of the genes, such as the frequency of the genes, their expression profiles, etc.

1a: Find your G/VOI (1.5 pt)

Instructions: Complete the following using your **G/VOI**

1. Go to the **UCSC Genome Browser**, center the track on your gene or variant of interest. If you are assigned a genetic variant, include the entire gene that the variant belongs to.
2. Locate the track showing the exons, introns, and direction of strands like the image below.
 - The direction of arrows indicates the 5' > 3' direction of a gene.
 - Each row of the blue genes is an isoform.



In the code cell below:

1. (0.5 pt) Store the location of the gene (transcript including UTR's) as shown in UCSC Genome Browser as a string in the variable `gene_location`.
2. (0.5 pt) Store the strand direction as a string in the variable `strand_direction`.
3. (0.5 pt) Store the number of exons as an integer in the variable `total_exon`.

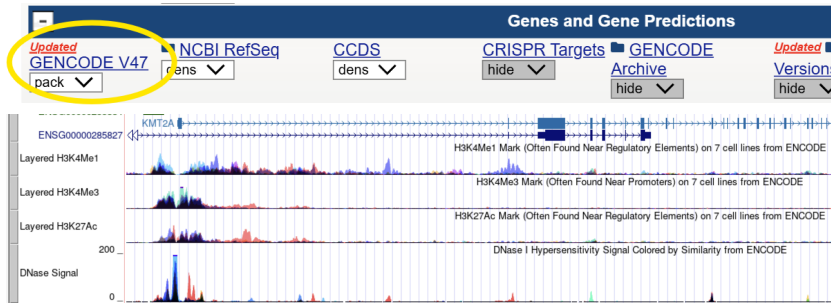
Here is an example for the gene KMT2A:

```
gene_location = "chr11:118436492-118526832"
strand_direction = "+"
total_exon = 36
```

```
In [ ]: gene_location = "chr19:11,089,463-11,133,820"
strand_direction = "+"
total_exon = 18
```

1b: Locate the regulatory elements using ENCODE tracks (2 pt)

1. Add the **ENCODE track** to your UCSC Genome Browser.



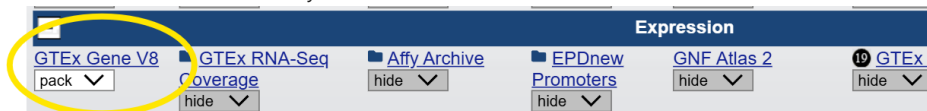
2. (2 pt) In 2-3 sentences, explain the locations of the peaks in the H3K Tracks and DNase I Hypersensitivity Track. Based on your knowledge of the gene structure, why do you see the peaks at these locations?

ANSWER: H3K acetylation and methylation peaks span exon 1 and into intron 1 with some peaks at exon 18. DNase Signal show highest sensitivity at and around exon 1. This makes sense because this is where UTR regions for expression would be located/open chromatin.

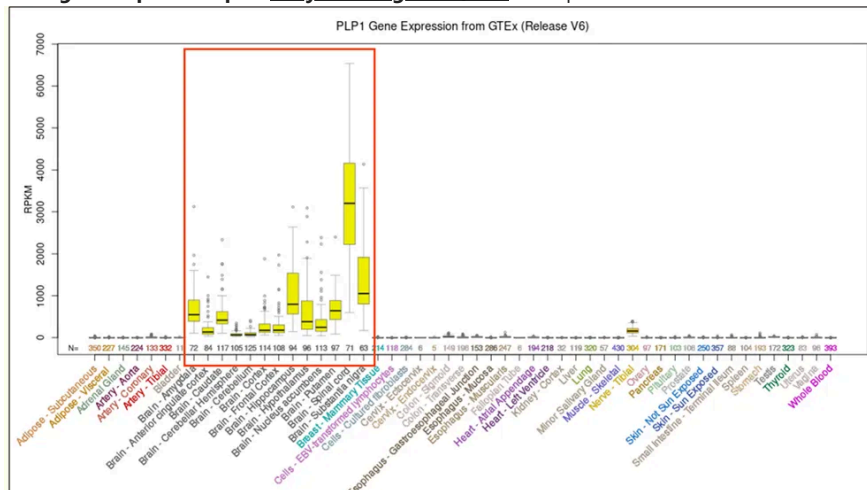
1c: Find tissue-specific gene expression using GTEx V8 track (2 pt)

Instructions:

1. Add the **GTEx Gene v8 track** to your UCSC Genome Browser.



2. Click on the added track to find the tissue-specific gene expression.
3. In two sentences, describe the tissues in which the gene expression are upregulated and downregulated based on the GTEx gene expression plot for your assigned G/VOI. Your plot should have the same x- and y-axes as the figure below.

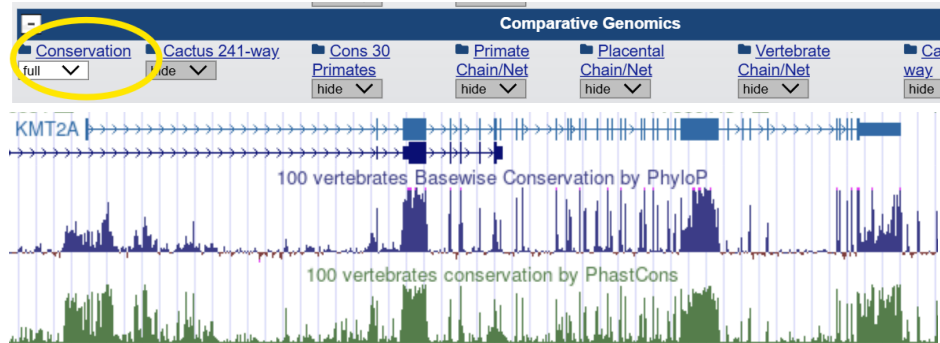


ANSWER: The tissues that are upregulated are Adrenal Gland, Lungs and Ovaries. The tissue that are down regulated is derived from the CNS (specifically brain)

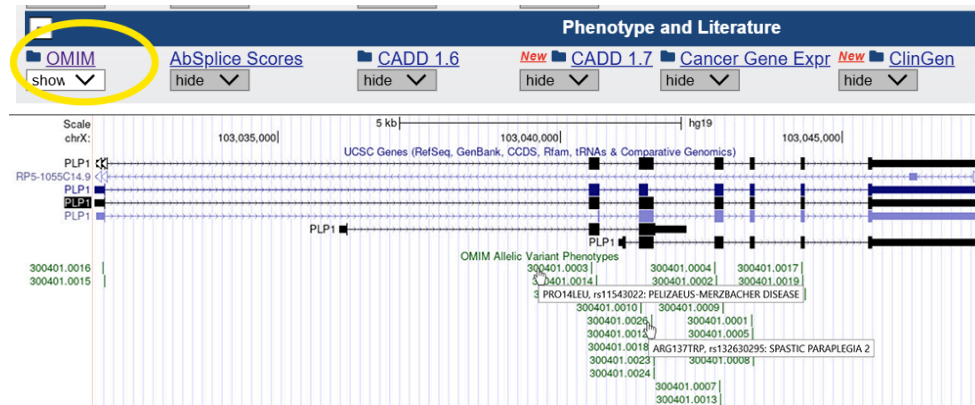
1d: Find conserved regions using conservation tracks (2 pt)

Instructions:

1. Add the **Conservation track** to your UCSC Genome Browser.



2. Add the **OMIM track** to locate the disease variants.



Provide the answer using your G/VOI in the markdown cell below:

- (0.5 pt) List one disease variant that lies in a region with high conservation score.
- (0.5 pt) List one disease variant that is in a region with low conservation score.
- (1 pt) What does the conservation score suggest about the rate of evolution for these disease variant

ANSWER

- High conservation region -> 606945.0015 FH FRENCH CANADIAN 2 (rs28942083)
- Low conservation region -> 606945.0055 FH AARHUS (rs397509365,rs875989944) *still relatively high conservation
- Inverse relationship between conservation score and rate of evolution, regions with high conservation show low rate of evolution

1e: Screenshot of your UCSC Genome Browser (1 pt)

Instructions:

- Attach a screenshot of your UCSC Genome Browser in the markdown cell below.
- Include all of the tracks mentioned in Exercise 1.

Exercise 3: Pathogenesis pathway (2 pt)

Based on the information in the first two exercises and accessible information in GeneCards and GeneReviews, describe the pathogenesis pathway of your assigned disease. Specifically, explain how your GVOI may contribute to the clinical phenotype of the disease. Cite the database(s) from which you found the information.

Using the example provided in exercise 2, when a deletion is present in the promoter region of LDLR results in no transcription of the LDLR gene, decreasing overall expression. The function of LDLR is as a surface receptor for endocytosis of low density lipoprotein/cholesterol (LDL) into the cell. The fact that organs with high flexibility (lungs) or steroid hormone production (adrenal gland and ovaries) express high levels of LDL indicates that decreased expression of LDLR results in lower results in less uptake of LDL by tissues and more LDL that is free floating